

hermes-agent Pull Requests 5761 and 5771

Summary of changes and review status · Repository: NousResearch/hermes-agent · Prepared 2026-10-07

Overview

Both pull requests are open, authored by jeannemtl, and target the **main** branch. They overlap heavily: PR 5771 contains both commits of PR 5761 plus one additional commit that edits an existing reference file. Both branch from the same base commit (972482e2).

	PR 5761	PR 5771
Title	feat(skills): add mathcode and manim-mathcode-explainer skills	fix(skills): fix manim-video skill ffmpeg rendering bugs
Branch	jeannemtl:feature/mathcode_manim_integration	jeannemtl:fix/manim-rendering-bugs
State	Open	Open
Commits	2	3 (the 2 from PR 5761 + 1)
Files	3 added	3 added, 1 modified
Lines	+699 / -0	+760 / -18
Labels	type/feature, P3, tool/skills, sweeper:*	type/bug, P3, tool/skills, sweeper:*
Link	github.com/NousResearch/hermes-agent/pull/5761	github.com/NousResearch/hermes-agent/pull/5771

PR 5761: add mathcode and manim-mathcode-explainer skills

Two commits adding three new files. Nothing is deleted. The commits are 2cd95c69 (mathcode skill) and c15d40f5 (explainer skill).

1. skills/autonomous-ai-agents/mathcode/SKILL.md (70 lines, new)

A new skill teaching the agent to install and drive MathCode, a terminal-based Lean 4 formalization CLI. Frontmatter: name: mathcode, version 1.0.1. It covers:

- Installation by cloning math-ai-org/mathcode and running its setup script.
- Switching from the default OpenAI Codex backend to Anthropic via the .env file (MATHCODE_USE_OPENAI=0, AUTOLEAN_USE_CODEX=0, ANTHROPIC_API_KEY).
- Always running with -p plus --dangerously-skip-permissions to bypass the interactive workspace-trust prompt.
- Writing unambiguous prompts. Example: "Jordan's Lemma" exists in both complex analysis and quantum computing, and a vague prompt produced a .lean file that conflated both and failed with a sorry placeholder.
- Splitting formalization from proof generation. Outputs land in LeanFormalizations/ as .lean and .eval.json files.
- A warning that heavy tactic proofs can run 50 to 60+ minutes and consume large amounts of API credit.

- Run in the foreground with a long timeout, or via the cronjob tool for true background work. Never use the terminal tool's background flag, which makes MathCode hang silently.

2. skills/creative/manim-mathcode-explainer/README.md (39 lines, new)

Installation guide listing the prerequisites: MathCode, Manim (pip), and ffmpeg. It explicitly says **do not install LaTeX**, because the pipeline uses Cairo Text () only and MathTex would crash. It also points to a verification script at scripts/setup.sh, which is not included in the PR.

3. skills/creative/manim-mathcode-explainer/SKILL.md (590 lines, new)

The actual pipeline for producing 2 to 3 minute "explain like I'm five" theorem videos. Its frontmatter declares name: manim-video (see review issues). Main sections:

- **Creative standard.** Narrative arc first, geometry before algebra, opacity layering, one cohesive visual language.
- **ELI5 vocabulary rules.** Only physical-object words (mirrors, flips, spins, doors). Terms such as self-adjoint, unitary, eigenvalue, or any mention of Lean are rejected. Includes the calibration history from the Jordan's Lemma video.
- **Proven 10-scene arc** with target durations: hook, core primitive, two primitives, reveal, rule, lemma statement, why always true, real-world stakes, mechanism, recap. Total about 2m10s.
- **Scene code templates** for each scene, with beat maps that tie animation timing to the narration.
- **Per-scene audio workflow** (required): generate narration per scene first, measure it, then write wait values to match.
- **Layout rules:** three-zone title-card-then-animation layout, no floating labels near moving objects, simple animations.
- **Critical pitfalls** learned in production, such as MoveAlongPath combined with .animate producing invisible motion, self.wait(0) raising, the Manim timestamp bug truncating concatenated video, and mixed-audio concat failures.
- **Full pipeline, render commands, performance targets, and color palette.**

The PR description says the workflow was validated end to end on macOS 15 (Apple Silicon M4, Python 3.9, Manim CE 0.18) with three videos: Jordan's Lemma, No-Cloning Theorem, and Pythagorean Theorem.

PR 5771: fix manim-video ffmpeg rendering bugs

Contains the same two commits as PR 5761, plus a third commit (327d563c) that modifies one existing file: skills/creative/manim-video/references/rendering.md (61 lines added, 18 removed). The motivation: three silent failures found while producing the Jordan's Lemma and No-Cloning videos with the pipeline from PR 5761.

What the third commit changes

- **Replaces the lossless stitch.** The old step used `ffmpeg -f concat -c copy`. Manim partial files carry non-monotonic DTS timestamps, so stream copy silently truncated output (74 s produced when 104 s was expected, no ffmpeg error). The new step re-encodes.
- **Adds a silent-audio pre-step.** Each scene gets a silent AAC track muxed in first, so scenes with and without audio can be concatenated together.
- **Adds a duration check.** Two `ffprobe` calls, one for container duration and one for the video stream, because they disagree when timestamps are broken. If they differ, re-encode.
- **Adds a tail-pad section.** When the voiceover is longer than the video, grab the last frame, loop it into a filler clip, and concatenate. The pad fps must match the scene fps (15 for -ql, 30 for -qm, 60 for -qh) or `concat` silently drops it.
- **Rewrites the voiceover mux.** Uses `-t "$AUDIO_DUR"` instead of `-shortest`, which clipped audio whenever the video ended first. The old per-scene audio concat example is removed.
- **Adds a note** claiming the `text_to_speech` tool always outputs .ogg regardless of the requested extension.
- Also removes a handful of blank lines in otherwise untouched sections below (manim.cfg, Sections, manim-voiceover).

Before and after, stitch step

```
# Before
ffmpeg -y -f concat -safe 0 -i concat.txt -c copy final.mp4

# After
for scene in Scene1 Scene2 Scene3; do
  ffmpeg -y -i "media/videos/script/480p15/${scene}.mp4" \
    -f lavfi -i anullsrc=channel_layout=stereo:sample_rate=44100 \
    -c:v copy -c:a aac -shortest "media/videos/script/480p15/${scene}_s.mp4"
done
ffmpeg -y -f concat -safe 0 -i concat.txt \
  -vf "fps=15" -c:v libx264 -preset fast -crf 22 -c:a aac stitched.mp4
```

Before and after, voiceover mux

```
# Before
ffmpeg -y -i final.mp4 -i narration.mp3 -c:v copy -c:a aac -b:a 192k -shortest final_narrated.mp4

# After
AUDIO_DUR=$(ffprobe -i voiceover.ogg -show_entries format=duration -v quiet -of csv="p=0")
ffmpeg -y -i full_video.mp4 -i voiceover.ogg \
  -c:v copy -c:a aac -map 0:v:0 -map 1:a:0 -t "$AUDIO_DUR" final_with_audio.mp4
```

Review status

Collaborator teknum1 reviewed both PRs with the same core objections. The triage bot GottZ noted that PR 5771 duplicates PR 5761 and recommended closing 5771 as a duplicate. The blocking items:

- 1 **Skill name collision.** The new explainer SKILL.md declares name: `manim-video`, identical to the existing bundled skill at `skills/creative/manim-video/SKILL.md`. The skill loader silently skips duplicates, so the new skill is never discoverable. It needs a unique name matching its directory.
- 2 **Missing file.** The README references `scripts/setup.sh`, which neither PR includes.
- 3 **Metadata non-compliance.** Descriptions run 271 and 452 characters against a 60-character limit. Frontmatter lacks the required author and platform fields. No tests exist at `tests/skills/test_<skill>_skill.py`. Reference: AGENTS.md lines 888 to 940.
- 4 **Incorrect TTS claim.** Verified against current `tools/tts_tool.py`: output is Opus .ogg only on voice-bubble platforms or when .ogg is explicitly requested, and MP3 otherwise. The "always .ogg" note is wrong.
- 5 **Incomplete fix.** Verified on current main: the old `-c copy concat` command still sits in the primary workflow at `skills/creative/manim-video/SKILL.md` line 185 (Step 4: Stitch). PR 5771 only fixes the `references/rendering.md` copy.

Suggested path forward

teknum1 recommends keeping PR 5771 as a focused fix that applies the corrected concat guidance to both Manim workflows (SKILL.md and `references/rendering.md`), with the TTS note corrected. The two new skills should move to a separate PR with unique directory-matching names, compliant frontmatter (short description, author, platform), the missing setup script, and skill tests. PR 5761 would then be closed as a duplicate.